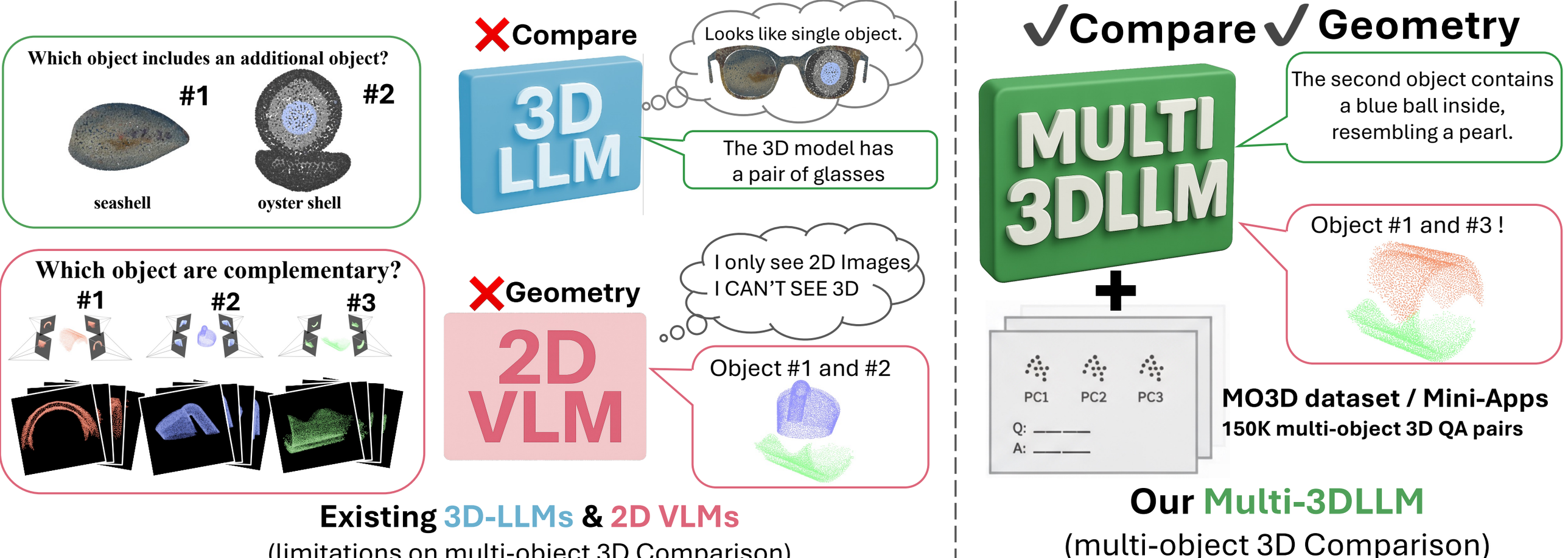
Which object includes an additional object?
#1
#2
seashell
oyster shell
Which object are complementary?
#1
#2
#3
✗Compare
3D LLM
Looks like single object.
The 3D model has a pair of glasses
✗Geometry
2D VLM
I only see 2D Images
I CAN'T SEE 3D
Object #1 and #2
Existing 3D-LLMs & 2D VLMs
(limitations on multi-object 3D Comparison)
✓Compare ✓ Geometry
MULTI 3DLLM
The second object contains a blue ball inside, resembling a pearl.
Object #1 and #3 !
PC1
PC2
PC3
Q: ____
A: ____
MO3D dataset / Mini-Apps
150K multi-object 3D QA pairs
Our Multi-3DLLM
(multi-object 3D Comparison)